

Harvey Barnes
10/8/2026

Exercise 1:

1a: for $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ and parameters $a = (a_1, a_2, a_3) \in \mathbb{R}^3$ $b \in \mathbb{R}$:

$$\tilde{p}(x, (a, b)) = 1 / (1 + e^{\{-(a_1 x_1 + a_2 x_2 + a_3 x_3 + b)\}}) = 1 / (1 + e^{\{-(a \cdot x + b)\}})$$

1b: for $x \in \mathbb{R}^n$, $a \in \mathbb{R}^n$, $b \in \mathbb{R}$:

$$\tilde{p}(x, (a, b)) = 1 / (1 + e^{\{-(a \cdot x + b)\}}), \text{ where } a \cdot x = \sum_{i=1}^n a_i x_i$$

Exercise 2:

2a:

$$L(A, b) = \sum_i \|y_i - Ax_i - b\|^2 \text{ residual } r_i = y_i - Ax_i - b \in \mathbb{R}^m \text{ so } L = \sum_i \|r_i\|^2$$

$$\partial L / \partial b = -2 \sum_i (y_i - Ax_i - b) = 0$$

$$b = (1/N) \sum y_i - A (1/N) \sum x_i$$

$$\|r_i\|^2 = \sum_j (y_{ij} - \sum_k a_{kj} x_{ik} - b_j)^2$$

$$\partial L / \partial a_{kj} = -2 \sum_i (y_{ij} - \sum_k a_{kj} x_{ik} - b_j) x_{ik} = 0$$

For $k = 1, \dots, M$ and $j = 1, \dots, n$

$$\sum_i (y_i - Ax_i - b) x_i^T = 0, \text{ i.e. } A \sum_i x_i x_i^T + b \sum_i x_i^T = \sum_i y_i x_i^T$$

$$\text{Sub } b = \bar{y} - A \bar{x}$$

$$A \cdot C_{xx} = C_{yx}$$

$$C_{xx} = \sum_i (x_i - \bar{x})(x_i - \bar{x})^T \text{ n*n matrix}$$

$$C_{yx} = \sum_i (y_i - \bar{y})(x_i - \bar{x})^T \text{ m*n matrix}$$

$$\text{Giving } A = C_{yx} C_{xx}^{-1}, b = \bar{y} - A \bar{x}$$

2b:

$$\text{Need } a \in \mathbb{R}^n, b \in \mathbb{R}, a \cdot x_i + b = y_i$$

$$Mz = y \text{ } z = (a, b) \in \mathbb{R}^{n+1}$$

$$M \text{ is } N \times (n+1) \text{ matrix rows } (x_i^T, 1)$$

Assumption is x_i are affinely independent

Rank $M = N$

$$\sum c_i (x_i, 1) = 0 \text{ then } \sum c_i = 0 \text{ and } \sum c_i x_i = 0$$

$$\text{Sub } c_1 = -\sum_{i \geq 2} c_i \text{ giving } \sum_{i \geq 2} c_i (x_i - x_1) = 0$$

Affine independence makes it linearly independent as well.

So $c_i = 0$ for $i \geq 2$ meaning $c_1 = 0$ so rows are independent

M has full row rank N so columns span $\mathbb{R}^N$ $Mz = y$ is solvable for all y . Since $n+1 > N$ there are infinitely many solutions giving a hyper plane $y = ax + b$ through the N points

2c:

Take 1d least squares line from a ($n = m = 1$) in case of part b, a is not unique so bound needs regression formula

$$a = \sum (x_i - \bar{x})(y_i - \bar{y}) / S \text{ and } x_i \text{ are not all equal}$$

$$\text{Since } \sum (x_i - \bar{x}) = 0$$

Y term drops out $a = \sum (x_i - \bar{x}) y_i / S$

Holds for $\tilde{a}$ with $\tilde{y}_i$ instead of y_i . x_i is same in both sets so S is same

$$a - \tilde{a} = \sum (x_i - \bar{x})(y_i - \tilde{y}_i) / S$$

With $|y_i - \tilde{y}_i| \leq \varepsilon$

$$|a - \tilde{a}| \leq (\varepsilon/S) \sum |x_i - \bar{x}| \leq (\varepsilon/S) \cdot N \cdot \max_i |x_i - \bar{x}|$$

Which leads to

$$|a - \tilde{a}| \leq C N \varepsilon \text{ and } C = \max_i |x_i - \bar{x}| / \sum_i (x_i - \bar{x})^2$$

C depends on x_i not on ε or y_i

Exercise 3:

3a: $c=2$,

$$(-1,4) \quad a-b+2 = 4, \quad a-b = 2$$

$$(1,-8) \quad a+b+2 = -8 \text{ so } a+b = -10$$

$$A = -4 \text{ and } b = -6$$

$$y = -4x^2 - 6x + 2$$

3b: line $y=x$ (0,0) and (2,2) third point (3,3.3)

$$4a+2b = 2, \quad 9a+3b = 3.3 \text{ so } a = .1 \quad b = .8$$

$$y = .1x^2 + .8x$$

$$n = 3, \quad \sum x = 5, \quad \sum y = 5.3, \quad \sum x^2 = 13 \quad \sum xy = 13.9$$

$$\text{Slope} = (3 \cdot 13.9 - 5 \cdot 5.3) / (3 \cdot 13 - 25) = 15.2/14 \approx 1.0857$$

$$\text{Intercept} = (5.3 - 1.0857 \cdot 5)/3 \approx -0.0429$$

$$Y = 1.0857x - 0.0429$$

3c:

Same first 2 points with third below

Same process gives

$$y = -.1x^2 + 1.2x$$

$$\text{And slope} = 1 - 3/35 = 0.9143, \quad \text{intercept} = 3/70 = 0.0429$$

$$y = 0.9143x + 0.0429$$

Moving by .6 in total has very different results

For point above its convex with $a = .1$, concave with $a = -.1$

Value at $x = 10$ is 18 for point above but 2 for point below for parabola

For the line fit it doesn't change as much with slope going from 1.086 to .914 and $x=10$ results in 10.8 and 9.8.

The line would be the better choice even though the parabola passes through each point it might not generalize well to new points. Meaning it's overfitting. The line on the other hand doesn't move much and doesn't memorize as much noise as the parabola does. It would generalize to new points better even if it didn't fit the training data as well as the parabola did.

Exercise 4: Logistic regression models $\log[p/(1-p)] = \beta_0 + \beta x$ and sigmoid is strictly increasing. So $p(x)$ can only go up or only go down in x or stay constant. If the true $p(x)$ rises then falls no B can match it. The fit then becomes a poor monotone one with B near 0. Leading to a single threshold on x not being able to separate the classes

Exercise 5:

1. Since $A1 = 2 \times 2$, having $R^3 \rightarrow R^2$ would require $(2 \times 2) (3 \times 1)$ which isn't possible
2. Same as above inner $a1$ can't multiply a (3×1)
3. **Defines $h(\vec{x}) : R^3 \rightarrow R^2$ as $A2(2 \times 3) \times (3 \times 1)$ produces a (2×1) matrix**
4. Inner $A1b$ gives (2×1) output but $A1x$ poses same problem as 1 and 2

Exercise 6:

$$d/du \log(u) = 1/u$$

$$d/du \log(1 - u) = -1/(1 - u)$$

$$\nabla_{\alpha} \log(p_0(s_i, \alpha)) = \nabla_{\alpha} p_0(s_i, \alpha) / p_0(s_i, \alpha)$$

$$\nabla_{\alpha} \log(1 - p_0(s_i, \alpha)) = -\nabla_{\alpha} p_0(s_i, \alpha) / (1 - p_0(s_i, \alpha))$$

$$\nabla_{\alpha} L(\alpha) = - (1/N) \sum_{i=1}^N \nabla_{\alpha} p_0(s_i, \alpha) / p_0(s_i, \alpha)$$

$$+ (1/M) \sum_{i=N+1}^{N+P} \nabla_{\alpha} p_0(s_i, \alpha) / (1 - p_0(s_i, \alpha))$$